Brayden Goldstein-Gelb

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Research Interests: Quantum algorithms and complexity theory, quantum error correction, mathematical structures in quantum computing, verifiable quantum advantage

Education

Boston University Boston, MA

PhD Computer Science 2026 – Present

- Advisor: Alexander Poremba

Brown University Providence, RI

Sc.B Computer Science (Honors), Sc.B Mathematics 2021 – 2025

- GPA: 3.91/4.00, Magna Cum Laude
- Honors Thesis: Distributed Quantum Signal Processing. (External Advisor: Yuan Liu, NC State; Reader: Brenda Rubenstein, Brown; Department Advisor: James Tompkin, Brown)

Research Experience

Undergraduate Researcher Providence, RI

Brown University, advised by Prof. Yuan Liu (North Carolina State University) Aug '24 – May '25

- Led a multi-institutional collaboration spanning MIT, Yale, NC State, and QuEra developing COMPAS, a distributed quantum algorithm for multivariate trace estimation in constant depth.
- Co-designed an architecture and algorithm to reduce classical and quantum communication.
- Performed simulations on HPC clusters and derived analytical bounds to demonstrate performance.

Undergraduate Research Intern, SULI Program Oak Ridge, TN

Oak Ridge National Laboratory, advised by Dr. Phil Lotshaw May '23 – Jul '23

- Investigated methods for solving constrained optimization problems using the Quantum Approximate Optimization Algorithm (QAOA).
- Developed a procedural method to translate constraint equations into mixing Hamiltonians.
- Identified a graph-theoretic interpretation of these Hamiltonians and proved connectedness for a broad class of constraints, establishing QAOA convergence guarantees.

Undergraduate Researcher, Brown Visual Computing Lab Providence, RI

Brown University, advised by Prof. James Tompkin. May '22 – May '23

- **Decoupled Style Descriptors Project:** Built an interactive application for handwriting synthesis via a recurrent neural network.
- **Augmented Reality Label Visualization Project:** Designed and implemented real-time algorithms for adapting augmented reality content to dynamic environments using Unity shaders.

Publications and Presentations

Publications

- **Brayden Goldstein-Gelb**, Kun Liu, John M. Martyn, Hengyun Zhou, Yongshan Ding, Yuan Liu. "COMPAS: A Distributed Multi-Party SWAP Test for Parallel Quantum Algorithms." In Proceedings of the 31st ACM International Conference on *Architectural Support for Programming Languages and Operating Systems (ASPLOS '26)*, Volume 2, pp. 426–441. arXiv:2511.23434
- **Brayden Goldstein-Gelb**, Phillip C. Lotshaw. "Convergence Guarantee for Linearly-Constrained Combinatorial Optimization with a Quantum Alternating Operator Ansatz." *ACM Transactions on Quantum Computing*. To appear. arXiv:2409.18829

Presentations and Talks

- COMPAS: A Distributed Multi-Party SWAP Test for Parallel Quantum Algorithms. *ASPLOS 2026*, Pittsburgh, PA, March 2026. Conference presentation.
- Environment Adaptive AR Label Visualization. *Brown Undergraduate CS Research Symposium*, Providence, RI, April 2023. Poster presentation.

Coursework

- **Computer Science:** Complexity Theory (grad., current), An Algorithmist's Toolkit (grad.), Design and Analysis of Algorithms, Theory of Computation, Algorithmic Machine Learning, Applied Cryptography, Systems, Formal Proof and Verification, Computer Vision
- **Quantum Computing:** Quantum Mechanics, Quantum Information, Probabilities in Quantum Mechanics
- **Mathematics:** Four semesters of Algebra (two undergrad, two grad), Real Analysis, Complex Analysis, Graph Theory, Information Theory, Number Theory, Optimization and Stochastic Calculus, Calculus on Manifolds, Statistical Inference, Linear Algebra

Teaching and Professional Experience

NewGrid Inc. Somerville, MA
Software Engineer II Jul '26 – Aug '26
Software Engineer I Aug '25 – Jun '26
Software Engineer Intern May '24 – Aug '24

- Developed new search capabilities and performance optimizations for power grid topology optimization algorithms.
- Built an external API and SDK for NewGrid's power grid simulation and analysis software.
- Automated the scraping, processing, and storage of power grid outage data, streamlining data collection for integration with NewGrid's database and enabling real-time monitoring for the service team.

Brown University Providence, RI
Grader, Algebra with Theory Jan '25 – May '25
Teaching Assistant, Theory of Computation Sept '24 – Dec '24

- Created problem sets and example solutions, assisted in grading, held weekly office hours.

Wyzant Inc. Providence, RI
Online Tutor May '21 – Aug '23

- Over 350 hours of tutoring experience, working with 50+ students ranging from middle schoolers to executives.
- Maintained a five-star rating, helping students build confidence and mastery in Python, Java, Calculus, and Algebra.

Projects

Hackathon Projects, MIT iQuHack, Brown Quantum Hackathon, Hack@Brown Spring '23 – Spring '25

- Won 2nd Place at Brown Quantum Hackathon
- Completed QuEra challenge at MIT's iQuHack quantum hackathon
- Created a space-themed game for Hack@Brown

Zero-Knowledge Proof for Graph 3-Coloring, Final Project, Applied Cryptography Spring '24

- Implemented an interactive zero-knowledge protocol to prove graph 3-colorability without revealing the coloring. *View project report.*
- Built a terminal-based application with verifier/prover roles communicating over TCP sockets.

Quantum Mechanics Proofs Library, Final Project, Formal Proof and Verification Fall '22

- Created a library in the Lean Theorem Prover to model quantum particles, measurements, and multi-particle states.
- Demonstrated the library's capabilities by constructing a formal proof of the no-cloning theorem.

More projects, additional details, and demos available at braydengg.com/projects.